

Alex Fang

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EDUCATION

Carnegie Mellon University | *Masters of Science (ML/NLP @ SCS LTI)*. GPA: 3.84/4.0 Pittsburgh, PA | December 2025

- **Coursework:** Generative AI, Advanced Natural Language Processing, Introduction to Machine Learning, Visual Learning & Recognition, Multimodal ML.
- **Activities:** NeuLab @ CMU

University of California, Los Angeles | *Bachelor of Science in Computer Science*. GPA: 3.8/4.0 Los Angeles, CA | June 2024

WORK EXPERIENCE

DatologyAI | *Member of Technical Staff* Redwood City, CA | May 2025 - Present

- Drove project on building customizable domain-specific and eval-targeted quality filters for web data; built an agent-driven, automated pipeline for quality filter design and iteration.
- Co-developed Luxical, an open-source lexical-dense text embedding library, for filter-based data curation; a Luxical-One + MLP quality classifier matches the FineWeb-Edu scorer's downstream LLM performance (3B, 60B tokens) while running 14x faster on CPU than the FineWeb-Edu scorer on GPU.
- Developed and led project for filtering synthetically generated code data using heuristic, model-based, and LLM-as-a-Judge approaches.

Murata Electronics | *Software Engineering Intern* Carrollton, TX | June 2023 - September 2023

- Designed and implemented an Expo React Native service application prototype which enables a carrier agnostic route to connect cellular IoT devices to cloud services via Azure IoT Hub.
- Aimed to facilitate cellular IoT end device application development and reduce time-to-market for clients.
- Visualized device JSON data routed through Azure IoT Hub event groups and controlled/configured a device from app using direct methods/device twin's desired properties using a Node.js and Express.js backend.

PROJECTS

Human Identity Preservation Metric for Image Editing | CMU Pittsburgh, PA | December 2025

- Developed HIPM, a Human Identity Preservation Metric for multi-turn image editing, addressing the gap that LPIPS, CLIPScore, and FID measure visual or semantic similarity but fail to capture whether a person's identity persists across edits.
- Built an end-to-end pipeline: designed two-axis LLM-as-judge rubrics (alignment + consistency), annotated 40K (source, edit, instruction) triplets via Gemini 2.5 Pro using stratified sampling and Flux.1 Kontext augmentation to correct bimodal score distributions, and LoRA-finetuned Qwen2.5-VL-7B-Instruct for joint score prediction.
- Achieved 0.53 / 0.37 Spearman correlation with human annotations on consistency and alignment respectively, which is 4-5x higher than LPIPS / CLIPScore baselines (~0.01-0.15) and outperforming Gemini 2.5 Pro (0.52 / 0.28).

Adventure Frames: Visualizing Text-Adventure Games Using Diffusion | CMU Pittsburgh, PA | December 2024

- Developed an approach to visualize frames of a text-adventure game with consistent scenes and entities across timestamps using a custom diffusion model fine-tuned using Textual Inversion.
- Generated a custom dataset of images for entities to fine-tune a stable diffusion model by generating text-adventure stories and extracting their entities using Gemini.
- Achieved an improvement in average consistency scores (scored using Inter-Annotator Agreement) from a baseline of 1 to 3.

End-to-End NLP System | CMU Pittsburgh, PA | October 2024

- Developed an end-to-end NLP system using a Retrieval-Augmented Generation (RAG) system with FAISS and BM25 vector retrieval methods, implemented using LangChain, roberta, Mistral, Llama, and a LoRA-finetuned Llama LLMs.
- Generated a fine-tuning dataset and testing dataset by generating question-answer pairs with context from a document dataset, compiled through web-scraping.
- Achieved 68% F1-score, 49% exact match, 67% precision, and 70% recall with the LoRA-finetuned Llama model using FAISS retrieval on the generated test dataset, showing significant improvement over Llama without RAG.

RESEARCH

Behavior Box | *Research Assistant under Dr. Graham Neubig* Pittsburgh, PA | September 2024 – May 2025

- Automatically identify types of text that language models underperform/over-perform (relative to other models) on, and describe these low-level "slices" in a human-interpretable way.
- Implemented a data visualization pipeline to investigate performance differences between Llama models on the Dolma dataset.

SKILLS

Languages and Frameworks: Python, C++, C, Bash, PyTorch, NumPy, LangChain.

Tools: AWS, Flyte, Spark, Ray, Databricks, HuggingFace, Microsoft Azure, Google Cloud Platform, Git.